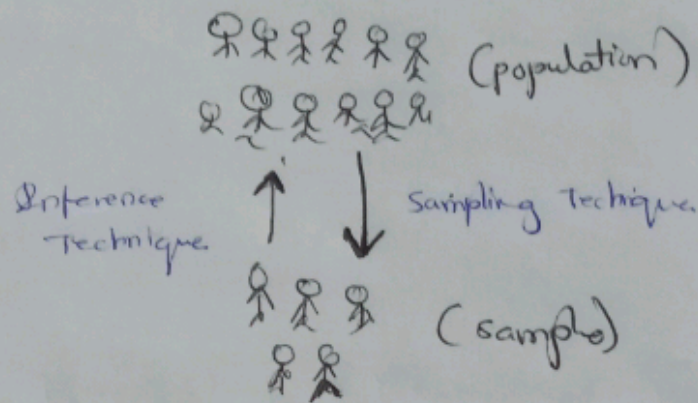


June-02

Statistics

Core Tools in Data Science

- Python, SQL, RPA, statistics = Backbone of any ML prediction model
- Understanding functions in python is essential.



population v/s Sample :-

population :- Entire group of individual / data.

Sample :- Subset of population used to infer about population.

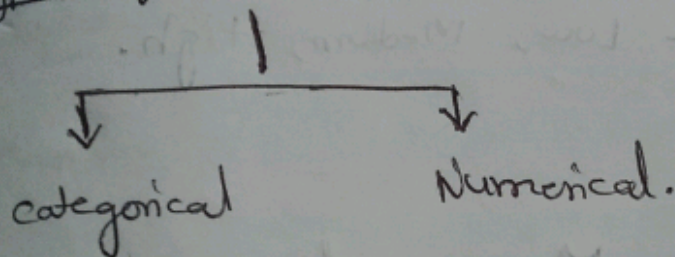
Sampling Technique :- Used to extract a sample from a population.

Inference Technique :- Used to draw conclusion from sample to population.

parameter :- ~~Numeric value~~ "parameter" is the characteristic of the population -
Numeric values that describe a population

Statistic :- Numeric value that describes a sample
(characteristic of sample data).

Types of Data:-



→ Numerical :- (Quantitative):

- Discrete
- Continuous

Numerical values you can measure or count.

Eg - Age, weight, marks.

Discrete :- Data that can take only certain values

Eg - no. of students

Continuous Data :- Data that can take any value within a range

→ Categorical Data (Qualitative):-

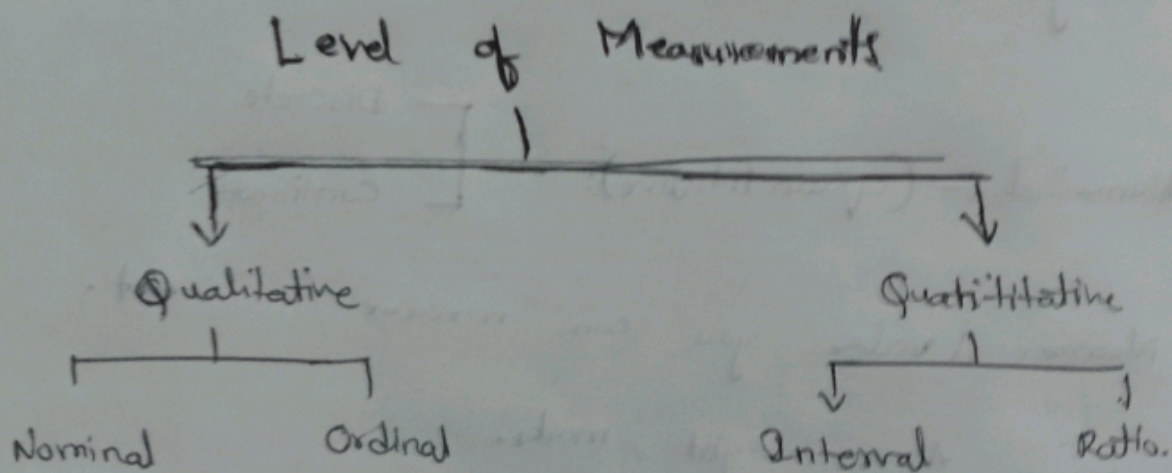
Data that represents categories or groups, and not numerical values

Subtypes — [Binary Data
Multi-class Data.

Binary Data: Only two categories (eg - Yes/No)

Multi-class Data: More than two categories

Eg - Low, Medium, High.



Nominal - categories, with no natural order

Eg - Gender, Blood type

Ordinal - Categories with a defined order but not evenly spaced

Eg - Rating (Good, Average, Poor)

Interval - Ordered data with equal but no true zero

Eg - Temperature in Celsius

Ratio - Like interval, but with a more meaningful zero

Eg - Height, weight, Age

Descriptive statistics:-

Mean - The mean is the sum of all values divided by the number of values.

It is also called the average.

- In excel, the mean is calculated by
= AVERAGE().

Median :- The median is the middle value of an ordered dataset. If the number of observations is even, it is the average of the two middle values.

- In excel, the median is calculated by:
= MEDIAN().

Mode :- The mode is the value that occurs most frequently in a dataset.

• In Excel, the mode is calculated by:

= MODE.SINGL() → returns one mode

= MODE.MULT() → returns an array with the modes. It is used when we have more than 1 - mode.

Variance :- Variance measures the average squared deviation of each number from the mean, indicating the spread of the data.

Standard Deviation :- (SD).

Standard deviation is the square root of variance and shows how much the data deviates from the mean.

Coefficient of Variation (CV)

Coefficient of Variation is the ratio of the standard deviation to the mean, often expressed as a percentage. It indicates relative variability.

Skewness :-

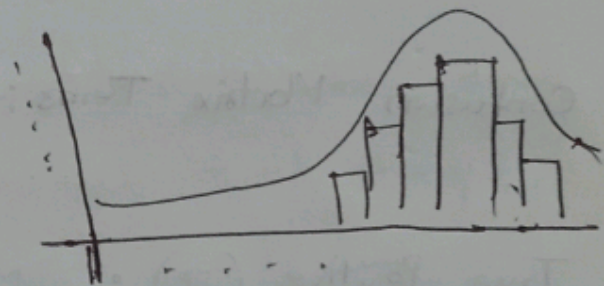
skewness measures the asymmetry of a distribution.
A distribution can be positively skewed (right-tailed)
negatively skewed (left-tailed) or symmetric
(zero skew).

Types

- Positive skew
- Negative skew
- Zero skew.



positive skew



Negative skew



zero skew.

Inferential statistics.

Inferential statistics involves making predictions or generalization about a population based on sample data.

Hypothesis Testing:-

Hypothesis testing is a method of making decisions using data by testing an assumption against an alternative Hypothesis.

Confusion Matrix Terms:-

True Positive (TP): Model correctly predicts positive cases.

False Positive (FP): Model incorrectly predicts positive when it's actually negative.

True Negative (TN): Model correctly predicts negative cases.

False Negative (FN): Model incorrectly predicts Negative when it's actually positive.

Regression Evaluation Metrics:-

R^2 (R-squared) - Proportion of the variance in the dependent variable that is predictable from the independent variable(s)

Adjusted R^2 - R^2 adjusted for the number of predictors; it penalizes unnecessary variables.

Random Sampling:-

Random sampling is a technique where every item in the population has an equal chance of being selected in the sample.

Outliers:-

Outliers are data points that differ significantly from other observations and can distort

statistical analyses.

- Affects mean significantly
- Less impact on median and mode
- Can affect ML model performance (uses models like Logistic Regression / Naive Bayes).

Exception Handling in python.

keyword -

try - Block where code that may cause an error is placed.

except - Block that catches and handles the error.

finally - code that runs no matter what (used for cleanup tasks).

Power BI use case:-

Power BI is a business intelligence tool used to visualize data for better insights. For example, it can be used to create interactive dashboards for 1000+ employees showing their performance and other metrics.